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Phosphate glasses for GRIN structures by ion exchange

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Abstract

Phosphate glasses were treated as a material for GRIN-lens production by silver-alkaline ion exchange. Glass durability (in salt melts) known to be a general obstacle for using these glasses in ion exchange technologies was found to profoundly increase in the sequence of meta ⇒ piro ⇒ orthophosphate. It is supposed to be due to the structure of phosphate glasses, which transforms in this sequence from 1D to 3D space arrangement of the glass-forming network. Ion exchange properties of these glasses were studied as well, and the maximal index variation in sodium–silver ion exchange was found to be 0.17.

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Keywords: Chemical durability; Ion exchange; Diffusion and transport; Glasses; Phosphates

1. Introduction

Today GRIN (graded-index) structures, like lenses, optical waveguides, diffractive gratings etc., are widely used in many fields of optics and optoelectronics [1,2]. Most of such structures are based on alkaline-containing glasses and produced by an ion exchange technique. It is because these glasses, being immersed in alkaline or silver salt melts, are capable of exchanging host univalent ions for ones in the salt melt, and this exchange results in increasing or decreasing glass refractivity in the ion exchanged layer. The magnitude of the refractive index variation (Δn_{max}), which can be achieved in ion exchange process, is that glass characteristic, which conditions limits of optical parameters of GRIN structures, which could be produced using a given glass. The bigger Δn_{max}, the wider these limits. In particular, numerical aperture (NA) of a GRIN lens is a linear function of $\Delta n^{1/2}$, where Δn is the index variation between optical axis of the lens and its lateral surface, with the Δn being limited by Δn_{max}. The value and sign of available Δn_{max}, in its turn, is strongly dependent on glass composition and types of ions participating in ion exchange.

Nowadays glasses and salt melts used for commercial production of converging GRIN lenses with a high NA (>0.5) contain thallium (highly toxic chemical element), and the process ensuring necessary value and sign of index variation is a direct ion exchange of host thallium ions for potassium ones (or other alkaline ions) contained in a salt melt. In particular, this process is used by the main GRIN lens producer – Nippon Sheet Glass Co., Ltd, who produces GRIN lenses with NA up to 0.58 that corresponds to $\Delta n \sim 0.1$. The only other non-toxic technology used in the production of converging GRIN lenses having $NA > 0.5$ (this corresponds to $\Delta n > 0.085$) is based on a double ion exchange [3] that makes the technology time consumable. This technology is used by GRINTECH GmbH. In a double ion exchange, the first ion exchange, which is the full ion exchange replacement of the host sodium ions with silver ones (this step is called the stuffing),

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is necessary to have a negative index change in the second one, which is the sodium ion exchange of silver-stuffed glasses. Note that negative sign of index variation is needed for the GRIN lenses to operate as a converging lens.

The main applied goal of this research is the development of new non-toxic glasses (and corresponding salt melts) to be used for producing high-NA converging GRIN lenses, that is, non-toxic glasses demonstrating a high negative value of $\Delta n_{\max}$ (0.1 and higher) in non-toxic direct ion exchange process.

In case of high negative $\Delta n_{\max}$, the alternative of the toxic thallium-alkaline ion exchange is known to be a non-toxic silver-alkaline one. To achieve a high value of $\Delta n_{\max}$, it is essential that glasses contain a great amount of ions capable of participating in ion exchange, for it is the amount of exchanged ions that conditions the magnitude of index variation $\Delta n_{\max}$. For a lens to be a converging one (negative value of dn/dr, where r is the lens radius), an initial glass should contain silver ions, which being ion exchanged with sodium ions from a salt melt give rise to just a negative value of dn/dr. It is well-known fact that silver ions are hardly incorporated by silicate glasses in great amount, excluding specific cases [4], but it is not so in case of phosphate glasses. It should be noted that, in addition to high $\Delta n_{\max}$, glasses used in ion exchange technologies must be of high chemical durability in the salt melts used. Phosphate glasses are known to show rather weak durability, and therefore, in spite of the possibility to achieve high values of $\Delta n_{\max}$ [5,6], today they are not used in GRIN lens production. However, by now nobody has performed a comprehensive study of chemical durability of silver-alkaline phosphate glasses in silver-alkaline salt melts. In this research we have studied just chemical durability of silver-alkaline phosphate glasses in silver-alkaline salt melts depending on general glass composition changing in the sequence of meta $\Rightarrow$ piro $\Rightarrow$ orthophosphate. The ion exchange properties of these glasses, in particular $\Delta n_{\max}$, have been characterized as well.

2. Experimental

About 100 glasses were synthesized with main components being P_2O_5 (25–65 mol%), Na_2O (25–40 mol%), Al_2O_3 (up to 36 mol%), B_2O_3 (up to 15 mol%), and Ag_2O (up to 30 mol%). Additional components, total content of which did not exceed 20 mol%, were RO, R_2O_3, RO_2, and R_2O_5, where R was one of the following chemical elements: Pb, Ba, Ca, La, Ti, Zn, and Nb. It was possible to sort all the glasses by their general composition (GC) as meta-, piro-, and orthophosphate, using Table 1. In this table we designated oxides of 2-valence and 4-valence metals as Me_2O_2 and Me_2O_4 rather than MeO and MeO_2, for the understanding of our classification to be more apparent.

We introduced more than three GCs indicated in Table 1, with only GCs 1, 2, and 3 being referred to exactly meta-, piro-, and orthophosphate compositions, correspondingly.

Table 1
General compositions (GC) of glasses

GC = 1 Meta -	GC = 2 Piro -	GC = 3 Ortho -
$Me_2O \cdot P_2O_5$	$2Me_2O \cdot P_2O_5$	$3Me_2O \cdot P_2O_5$
$Me_2O_2 \cdot 2P_2O_5$	$2Me_2O_2 \cdot 2P_2O_5$	$3Me_2O_2 \cdot 2P_2O_5$
$Me_2O_3 \cdot 3P_2O_5$	$2Me_2O_3 \cdot 3P_2O_5$	$3Me_2O_3 \cdot 3P_2O_5$
$Me_2O_4 \cdot 4P_2O_5$	$2Me_2O_4 \cdot 4P_2O_5$	$3Me_2O_4 \cdot 4P_2O_5$
$Me_2O_5 \cdot 5P_2O_5$	$2Me_2O_5 \cdot 5P_2O_5$	$3Me_2O_5 \cdot 5P_2O_5$

Note: Me is any chemical element mentioned in the text, except P.

A glass was considered belonging to a certain GC if the content of P_2O_5 was high enough to be in accordance with the ratios indicated in Table 1 for all other glass components. Classes of GC equal to, say, 1.5 corresponded to the glass compositions lying between GCs 1 and 2. Such half-integer GCs can be calculated by the expression $GC = \frac{\sum_i m_i X_i}{Y}$, where m_i and X_i are the valence and the molar content of oxide of metal Me_i in the glass ($Me \neq P$), and Y is the molar content of P_2O_5. (To use this expression, present oxides like MeO as Me_2O_2.) The GCs show the total number of valence bonds of all the metals bonded with one phosphorous through non-bridging oxygen.

The glasses (150 g of each) were melted at 1250–1500 °C with stirring for 2 h in silica glass crucibles. In batch preparation only chemically pure and high-purity grade reagents were used. The melts were poured out into a carbon mould and annealed for 2 h at the temperature corresponding to the glass viscosity equal to about 10^{12} Pa s.

Chemical durability of glasses was firstly checked using only sodium-contained glasses, and then most promising glasses, that is, glasses demonstrated good durability, were synthesized with partial or full replacement of sodium for silver, and the durability of these glasses were checked again. Samples for this ion exchange were cubes about $15 \times 15 \times 15\ mm^3$, the cube faces being polished. The durability was checked by ion exchange in different salt melts, such as salt melts of $NaNO_3$, $AgNO_3$, and mixtures of $NaNO_3$ and $AgNO_3$ with Na_2SO_4 NaCl, Na_2MoO_4, and $ZnCl_2$. Temperatures used in ion exchange were 350, 410, 490, and 540 °C, and standard duration of ion exchange treatment was 24 h. Chemical durability of the glasses was evaluated and classified by the following features; class 1 – very bad (crusty, surface destroyed layer, essential sample dissolution), class 2 – not good enough (surface milky film and weight losses more that a few percents), class 3 – good (no visible damages and a few weight losses, thin surface film), and class 4 – very good (no visible damages and no weight losses).

To determine the value of $\Delta n_{\max}$, ion exchange treatments of glass samples were performed in a silver nitrate salt melt at 350 °C (below glass transition) for 30 min, only sodium-contained glasses being used. The samples were slabs ($15 \times 20 \times 1\ mm^3$) with high-quality polished big faces. The ion exchange resulted in optical waveguides. Effective indices of the formed waveguides were measured by the prism-coupling technique with He–Ne laser ($\lambda = 0.63$ μm), and then the found indices were used to calculate refractive index profiles. In these calculations the algorithm proposed by Heidrich and White [7] for the solution of the inverse WKB-problem was used. At the same time, samples of silver-containing glasses were ion exchanged in a sodium nitrate salt melt at the same temperature for 2–4 days. Such a long ion exchange treatment led to rather deep index profiles with smaller index gradient at the sample surfaces and allowed estimating the achieved index variations with a refractometer by the measurements of glass index before and after ion exchange.

3. Results

It was not found any essential difference in the influence of the salt melt compositions used in this research on glass durability. The most critical factor was found to be the temperature of ion exchange treatment. There were no glasses, which demonstrated good durability at 490 and 540 °C, and only part of the designed glasses proved to be of good durability at temperatures below 410 °C only. The summarized results of studying chemical durability at 350 and 410 °C are presented in Fig. 1. One can see that in the sequence of meta ⇒ piro ⇒ orthophosphate chemical durability of phosphate glasses increases.

Fig. 2 depicts examples of index variation profiles formed at 350 °C for 30 min in several sodium-containing glasses by silver ion exchange. Compositions of these glasses are presented in Table 2. One can see that the magnitude of index change vary from 0.13 to 0.17, the index change being positive. The 2–4 day sodium ion exchanges performed in silver-containing glasses gave rise to a negative index change exceeding −0.1.

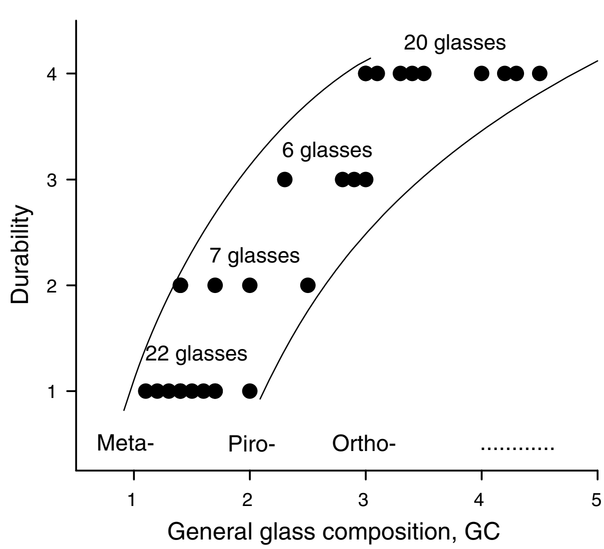

Fig. 1. Relation between chemical durability in salt melts and general glass composition (GC).

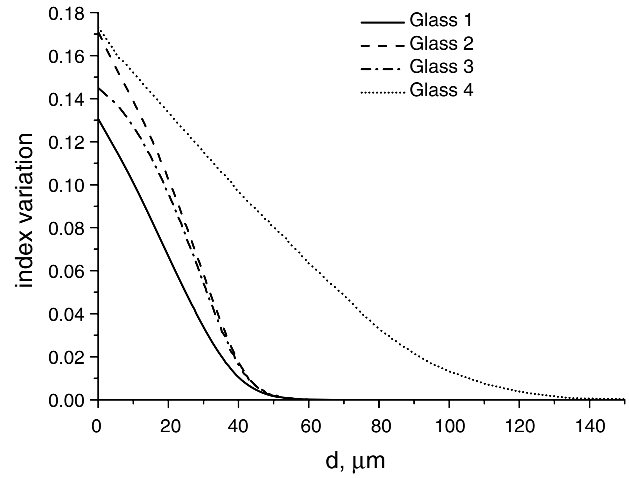

Fig. 2. Index profiles formed by silver ion exchange at 350 °C for 30 min in several sodium-containing glasses (see Table 2); d is the distance from the sample surface.

Table 2
Compositions (in mol%) of the glasses presented in Fig. 2

	P_2O_5	Na_2O	Al_2O_3	B_2O_3	ZnO	TiO_2	Nb_2O_5
Glass 1	31.4	31.4	31.4	5.8	–	–	–
Glass 2	33	30	37	–	–	–	–
Glass 3	34	33	25	–	8	–	–
Glass 4	41	38	–	2	–	4	15

4. Discussion

We interpret the observed rise of chemical durability of the designed phosphate glasses in the sequence of meta ⇒ piro ⇒ orthophosphate as the result of the fact

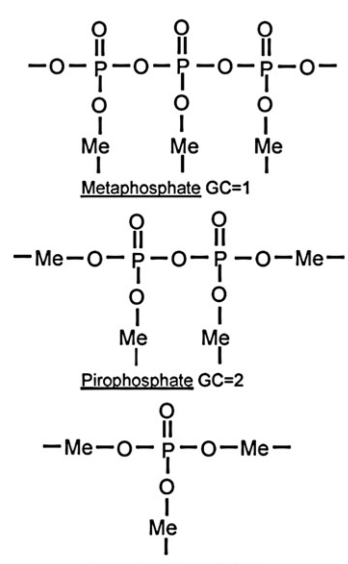

Fig. 3. Shematic presentation of the transformation of the structure of phosphate glasses from 1D to 3D space arrangement of the glass-forming network in the sequence of meta ⇒ piro ⇒ orthophosphates of bivalent metal Me.

that in this sequence structure of phosphate glasses transforms from 1D to 3D space arrangement of the glass-forming network. Meta-, piro-, and orthophosphates of some metal Me, say 2-valence Me^{2+}, can be formally presented as it is depicted in Fig. 3. In continuous phosphate network such metal ions play role of 'a tailor' which sews phosphate chains. The higher amount of valences of metal ions per one valence of a phosphorus ion, the higher amount of 'bridges', which could sew neighbor phosphate chains and transforms glass structure from 1D to 3D. It is this ratio that is the factor GC in Fig. 1. That 1D structure dissolves faster as compared with 3D is evident from steric reason.

From Fig. 2 one can see that the designed glasses demonstrate index profiles differing in their index variation, diffusion depths and shapes. The latter is very important because the shape of index profile is conditioned by the diffusion nonlinearity of the glass, i.e. concentration dependence of diffusion coefficient, and knowledge on how glass components influence this nonlinearity gives possibility to vary this nonlinearity and to tune glass composition in such a way to produce high quality GRIN lenses using minimal number of ion exchange steps [8].

Finally, we emphasize that most promising compositions of sodium-contained glasses (with good durability, faster diffusion, and bigger $\Delta n_{\max}$) were successfully used to synthesize glasses where sodium was replaced (up to 30 mol%) with silver. The magnitude $\Delta n_{\max}$ for studied silver-contained glasses corresponds to maximal NA ~ 0.56, and the variety of index profile shapes gives a good chance to produce GRIN lenses using chemically durable glass. Such glasses can be used in producing GRIN lenses not using toxic chemical elements.

5. Conclusion

A set of phosphate glasses with great amounts of either sodium or silver ions (about 100 glasses) has been tested in ion exchange experiments to find such glass compositions, which would have properties suitable for producing high-NA GRIN lenses in direct ion exchange process (without stage of the suffing). Two main properties have been studied, namely: chemical durability in salt melts and maximal index variation achievable in ion exchange. It has been found that chemical durability of these glasses increases in the sequence of meta $\Rightarrow$ piro $\Rightarrow$ orthophosphate, and orthophosphate glass are capable of incorporating up to 30 mol% Ag_2O while melted. Silver-containing glasses of general compositions corresponding to orthophosphates have demonstrated just suitable chemical durability and high negative index change (>0.14) in sodium ion exchange that makes the designed glasses promising for high-NA GRIN lens production.

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